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Locality Optimization Refinement with Deformation for Shape Matching via Functional Maps

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Abstract

Pointwise map recovery, a critical component in functional maps, is of paramount importance for nonrigid shape matching, but is inherently inaccurate due to the lack of geometric consistency. To this end, we propose a novel and effective framework, termed *Locality Optimization Refinement with Deformation* (LORD), which firstly integrates both local and global extrinsic constraints into functional space for precise pointwise maps. Specifically, observing local consistency during deformations, we innovatively devise a mathematical model with a linear closed-form solution leveraging neighborhood support, thus ensuring continuity of pointwise maps. To address the limitations of local constraints in contextual representation, we further introduce the kernel trick in non-Euclidean spaces to design probabilistic models based on functional deformation that enhance global smoothness. Lastly, we propose outlier refinement using Locally Linear Embedding and estimated functional deformation within an adaptive spectral similarity space, making it suitable for various types of shapes. Our comprehensive geometric constraints eliminate functional mapping ambiguity and significantly improve accuracy with linearithmic complexity. Extensive experiments conducted on different public benchmarks demonstrate that our LORD outperforms state-of-the-art techniques in nonrigid shape matching tasks, and verify its generalizability and effectiveness.

Keywords Shape matching · Functional maps · Pointwise conversion · Outlier · Local consistency · Functional deformation

1 Introduction

Constructing correspondences or maps between two shapes is a fundamental and longstanding problem in computer vision and graphics (Van Kaick et al., 2011; Sahillioğlu, 2020), with broad applications ranging from shape analysis (Hartman et al., 2023), deformation transfer Sumner & Popović (2004), to texture mapping (Ezuz & Ben-Chen, 2017), among others. Unlike rigid alignment with simple parametric modeling (Besl & McKay, 1992), matching nonrigid shapes as a generic case presents an intractable challenge due to its high complexity and unknown outliers (Deng et al., 2022).

Owing to the approximately isometric nature of deformations, such as pose changes, in real-word scenarios, there have been significant research interests in estimating near-isometric maps between nonrigid shapes in recent years (Sahillioğlu, 2020). Among the multitude of strategies addressing this issue, the functional maps framework, initially proposed by Ovsjanikov et al. (2012), stands out as an exemplary technique due to its efficiency. By generalizing the notion of mapping to establish correspondences between real-valued functions rather than points on shapes, the functional map framework provides a mapping representation based on the spectral space of each shape, such as the eigenfunctions of the Laplace-Beltrami (LB) operator. This allows natural constraints on a map, such as descriptor preservation and operator commutativity, to become linear. Moreover, algebraic operations on the maps, such as summation and composition, support the transfer of shapes accordingly. For example, the orthogonality of functional map matrices is associated with local volume preserving maps, the commutativity of Laplacian operator corresponds to near isometries, and conformal maps preserve functional inner products (Donati et al., 2022). These properties enable

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the functional map framework to serve as an efficient and flexible shape matching strategy. For a detailed explanation, readers are encouraged to refer to the introductory course (Ovsjanikov et al., 2016).

Numerous research following functional maps framework have emerged, extending to various tasks (Rustamov et al., 2013), correspondence models (Rodolà et al., 2017; Kovnatsky et al., 2013), and incorporating desirable constraints using linear algebraic formulations (Nogneng & Ovsjanikov, 2017; Eynard et al., 2016; Ren et al., 2018). Several works have been proposed to improve the effectiveness of functional maps within the spectral domain (Ren et al., 2018; Melzi et al., 2019b; Colombo et al., 2023; Hu et al., 2021; Cao et al., 2023). These methods primarily enhance the representations of features, but their recovery of pointwise maps relies on the ICP-based iterative strategy (Besl & McKay, 1992) introduced in the original work (Ovsjanikov et al., 2012), *i.e.*, a Nearest Neighbor (NN) search between eigenfunction matrices. Because the theoretical justification for pointwise conversions is inherently non-rigorous, precise alignment at fine scales is hindered, and properties such as continuity and smoothness are lacking.

The ill-defined issue of pointwise conversions within the functional maps framework has been specifically addressed in some works (Rodolà et al., 2015; Rodola et al., 2017; Ren et al., 2021; Pai et al., 2021; Fan et al., 2022). Overall, current techniques for pointwise mapping recovery face significant limitations, either lacking sufficient precision (*e.g.*, NN) or suffering from time and memory overheads (*e.g.*, assignment solvers). Further in-depth investigation is warranted to address these challenges.

In this paper, we propose a novel and efficient framework, named *Locality Optimization Refinement with Deformation* (LORD), for recovering accurate pointwise maps with significant continuity and smoothness. Given noisy pointwise maps, we design a two-step strategy to refine point-to-point correspondences, *i.e.*, distinguishing inliers (correct correspondences) and refining outliers (erroneous correspondences). Firstly, by observing local consistency across various deformations, we establish a mathematical model with closed-form solutions that possess linear complexity to efficiently discriminate inliers. Considering the limitations of local constraints, such as the contamination of continuous outliers, we introduce a global probabilistic model based on functional deformation. This model avoids local optima by utilizing local consistency-based matching results as initialization, obtaining reliable inliers through the Expectation-Maximization (EM) algorithm. In the second step, we utilize the reliable correspondences of inliers and the estimated functional deformations to refine outliers. We construct neighborhoods of outliers from inliers in the source shape and estimate weights based on Locally Linear Embedding (LLE). Then we search for correspondences of outliers in the spectral domain that satisfy both LLE and deformation constraints as the re-matching results. Moreover, the hyperparameters are set adaptively, ensuring applicability to shape data of varying resolutions and types. Visual comparisons with the recent learning-based state-of-the-art UNS-FMNet (Colombo et al., 2023) and baseline MWP (Hu et al., 2021) are illustrated in Fig. 1.

Specifically, the main contributions of this paper can be summarized as follows:

- We propose a novel and effective two-step framework for recovering pointwise maps, *i.e.*, distinguishing inliers and refining outliers, which embeds extrinsic topological constraints into functional space for precise point correspondences.
- Observing the local consistency of nonrigid deformations, we propose a concise mathematical model with a closed-form solution of linear complexity to distinguish inliers, with LLE is utilized to represent local topology during outlier refinement step.
- To address the insufficient contextual representation of local constraints, we introduce the kernel trick in non-Euclidean space to establish a functional deformation-based probabilistic model, achieving global smoothness.
- We conduct extensive experiments on multiple public pipelines, encompassing various types of shapes. We compared our LORD with other state-of-the-art methods and demonstrated its superiority.

Preliminary conference version of this paper was previously presented in LOPR (Xia et al., 2024), and our LORD is a comprehensive extension of LOPR, which focused solely on local consistency. This limited consideration leads to errors when encountering continuous, localized outliers. Furthermore, the inadequate utilization of contextual information constrains the robustness to complex deformations.

To address these issues, we introduce a global probabilistic model based on functional deformation to comprehensively describe the transformation between nonrigid shapes. Leveraging kernel techniques from non-Euclidean domains, we construct a probabilistic model using truncated bases with smoothness constraints and estimate the underlying deformation using the EM algorithm. In the subsequent outlier correction step, the estimated deformation serves as a crucial reference to recover pointwise maps more accurately. Briefly, we innovatively embed both local and global geometric constraints in the functional space to ensure the geometric consistency of pointwise maps. To clearly illustrate the differences between our LORD and the previous LOPR, a visual example is presented in Fig. 2. Compared to LOPR, LORD introduces functional deformation, which better distinguishes correct correspondences, thereby yielding smoother refined matching. Given the same number of

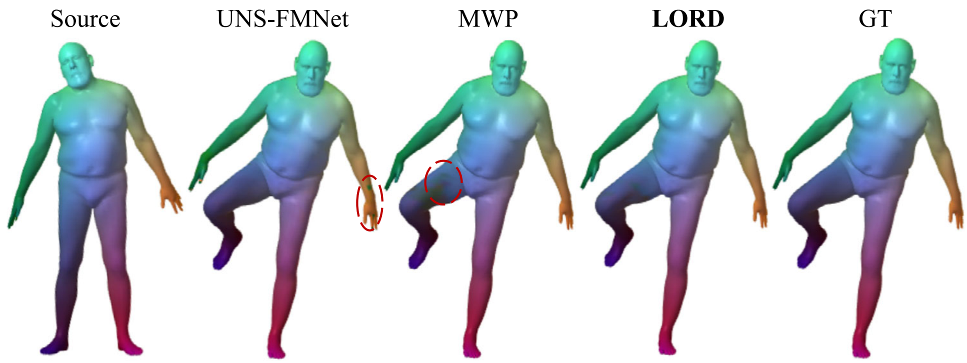

Fig. 1 An illustration using color transfer, where our LORD is compared with the learning-based state-of-the-art UNS-FMNet (Cao et al., 2023) and our baseline MWP (Hu et al., 2021) on the classic FAUST dataset (Bogo et al., 2014). Notably, our LORD demonstrates more accurate pointwise maps

iterations, LORD yields fewer incorrect correspondences and achieves an approximate 19.6% decrease in average geodesic error in this example.

For the experiments, we employ the enhanced method LORD to compare with state-of-the-art approaches on more challenging publicly available datasets (SHREC'19 (Melzi et al., 2019a) and DT4DM (Li et al., 2021)), demonstrating the superiority of the proposed method. Finally, we conduct further analysis of LORD to validate its stability in pointwise conversion applications and reveal the effectiveness of each component.

The rest of this paper is organized as follows. Section 2 describes related research works. In Sect. 3, we revisit the functional maps framework. Section 4 details our method. In Sect. 5, we apply our method to several public datasets for evaluation. The concluding remarks are summarized in Sect. 6.

2 Related Work

Shape matching is a well-researched field, and interested readers can refer to recent surveys (Van Kaick et al., 2011; Biasotti et al., 2016; Deng et al., 2022; Sahillioğlu, 2020) for a comprehensive exploration. Below, we review works relevant to ours.

2.1 Metric Space Methods

Most early nonrigid shape matching methods focused on directly discovering point-to-point correspondences between shapes while preserving the metric space as well as possible. Multidimensional scaling methods (Borg & Groenen, 2005; Bronstein et al., 2006) embed shapes into an intermediate space where nonrigid objects can be viewed as rigid. Classical ICP (Besl & McKay, 1992) can be used to perform matching in this process, as seen in Elad and Kimmel (2003) and follow-up work (Lipman & Funkhouser, 2009). A straightforward idea is to embed one shape into another (Bronstein et al., 2006), with its solver based on the Gromov-Hausdorff distance framework (Mémoli & Sapiro, 2005). Following this, the Quadratic Assignment Problem (QAP) can be adopted to incorporate geodesic distance preservation into a quadratic objective, which is then optimized over permutation matrices (Bernard et al., 2018). In brief, these methods directly establish point-to-point correspondences between surfaces by minimizing explicit energy terms, similar to Huang et al. (2008); Ovsjanikov et al. (2010). The main limitation of these methods is the complexity of combinatorial problems, which leads to a significant computational burden. Several relaxation-based methods have been designed to address this issue (Leordeanu & Hebert, 2005; Solomon et al., 2012), but they are not applicable to complex meshes.

2.2 Functional Maps Methods

The functional map framework (Ovsjanikov et al., 2012) involves estimating linear transformations between the functional spaces on shapes, using a reduced eigenfunction basis. It has inspired numerous works, including partial shape matching (Rodolà et al., 2017; Litany et al., 2017), matrix completion with manifold structure (Kovnatsky et al., 2015), discrete optimization (Ren et al., 2021), and alternative functional space representations (Kovnatsky et al., 2013; Wang et al., 2018).

Some representative instances are proposed to improve functional maps. For instance, BCICP (Ren et al., 2018) introduces an orientation-preservation energy to address intrinsic symmetry ambiguity, ZoomOut (Melzi et al., 2019b) improves the accuracy of iterative estimation through spectral upsampling, and PC-GAU (Colombo et al., 2023) devises a new functional basis, Principal Component of a Dictionary, to address intrinsic limitations of LB. Smooth Shells (Eisenberger et al., 2020) combines intrinsic and extrinsic information in this product space for robust matching. In particular, MWP (Hu et al., 2021) preserves multiscale spectral manifold wavelets for remarkably efficient matching performance. Moreover, the functional map framework has been combined with deep learning (Litany et al., 2017; Halimi

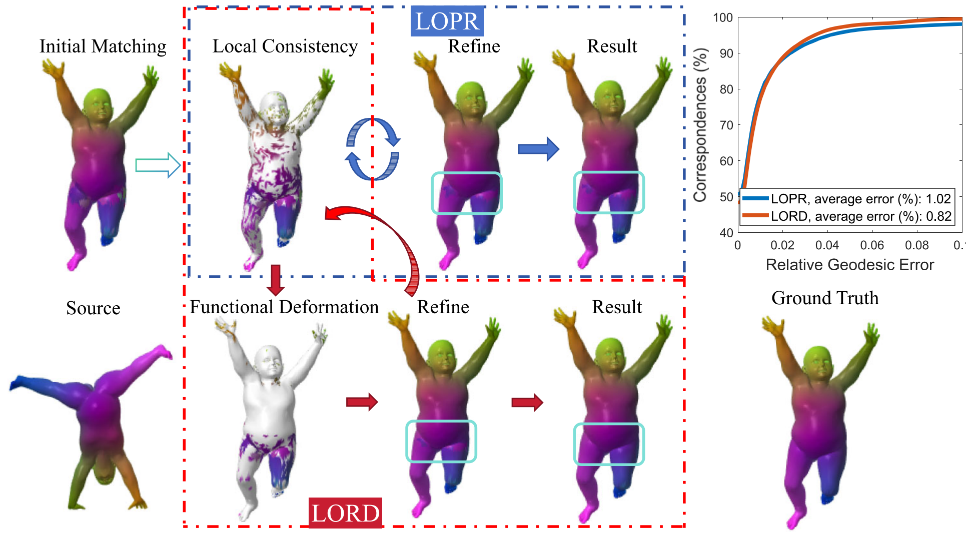

Fig. 2 Comparison between our LORD and its previous version, LOPR, for transferring source color to a target shape. Given a coarse initial matching, the blue boxes and arrows illustrate the LOPR pipeline, while the red boxes and arrows donate the LORD pipeline. White regions represent inliers identified by either local consistency or functional deformation. The cyan boxes highlight the regions prone to correspondence errors

et al., 2019). The state-of-the-art UNS-FMNet (Cao et al., 2023) couples functional maps with point maps during learning by unsupervised manner. However, data-driven methods lack generalizability due to the limitation of training data. The major challenge in this domain is extracting pointwise maps from functional maps (Rodolà et al., 2015), due to the limited expressiveness of functional spaces on truncated bases.

2.3 Functional Map Conversion

Most relevant to our work are studies that directly address the recovery of point-to-point correspondences from functional maps, *i.e.*, pointwise conversion, which is a significant step in all methods based on functional maps (Rodolà et al., 2015; Rodola et al., 2017; Ezuz & Ben-Chen, 2017). Notably, PMF (Vestner et al., 2017b) and KernalMatching (Vestner et al., 2017a) both utilize kernel density estimation based on geodesic distances, albeit with considerable computational burdens. Fast Sinkhorn Filters (Pai et al., 2021) introduces a matrix scaling framework derived form computational optimal transport, yet it is limited to spectral embedding alignment, leaving room for improvement. COMB (Roetzer et al., 2022) devises a scalable combinatorial solver but suffers from high computational burden. Additionally, GCPD (Fan et al., 2022) achieves extrinsic registration based on the classic CPD (Myronenko & Song, 2010) but still relies on functional maps for initialization. EDEO (Magnet et al., 2022) emphasizes the smoothness of pointwise maps using Dirichlet energy. Despite significant progress, the decoupling of conversion between pointwise correspondences and functional maps remains a major difficulty.

3 Functional Maps Revisited

Given a shape modeled as a smooth, compact, two-dimensional manifold $\mathcal{M}$ with area element $d\mu$ embedded in $\mathbb{R}^3$, the space of square-integrable functions on the manifold $\mathcal{M}$ is denoted by $\mathcal{L}^2(\mathcal{M}) = \left\{f : \mathcal{M} \to \mathbb{R}, \int_{\mathcal{M}} f(x)^2 d\mu < \infty\right\}$. With the symmetric Laplace-Beltrami operator $\Delta_{\mathcal{M}}$ providing Fourier analysis on the manifold $\mathcal{M}$, an eigen-decomposition $\Delta_{\mathcal{M}}\phi_i = \lambda_i\phi_i$ for $i \geq 1$ exits with eigenfunctions $\{\phi_i\}_{i\geq 1}$ as an orthonormal basis and eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots$ of $\mathcal{L}^2(\mathcal{M})$. In this case, any function $f \in \mathcal{L}^2(\mathcal{M})$ can be represented as $f(x) = \sum_{i\geq 1} \langle f, \phi_i\rangle_{\mathcal{M}} \phi_i(x)$, where $\langle\cdot,\cdot\rangle_{\mathcal{M}}$ denotes the inner product on the manifold $\mathcal{M}$.

3.1 Functional Correspondence

As proposed by the original work (Ovsjanikov et al., 2012), the shape correspondences can be obtained by transferring functions between manifolds. Given a pointwise map $T : \mathcal{M} \to \mathcal{N}$ and a functional map $T_F : \mathcal{L}^2(\mathcal{M}) \to \mathcal{L}^2(\mathcal{N})$ from a manifold $\mathcal{M}$ to another manifold $\mathcal{N}$, the image of any function $f \in \mathcal{L}^2(\mathcal{M})$ is defined as $T_F(f) = f \circ T^{-1}$. Considering two orthonormal bases $\{\phi_i^{\mathcal{M}}\}_{i\geq 0}$ and $\{\phi_i^{\mathcal{N}}\}_{i\geq 0}$ of $\mathcal{L}^2(\mathcal{M})$ and $\mathcal{L}^2(\mathcal{N})$, respectively, the functional image is represented as:

$$T_F(f) = \sum_j \sum_i \left\langle f, \phi_i^{\mathcal{M}}\right\rangle_{\mathcal{M}} \underbrace{\left\langle T_F\left(\phi_i^{\mathcal{M}}\right), \phi_j^{\mathcal{N}}\right\rangle_{\mathcal{N}}}_{c_{ji}} \phi_j^{\mathcal{N}}. \quad (1)$$

We denote the matrix $\boldsymbol{C} = [c_{ji}] \in \mathbb{R}^{k\times k}$, which maps the first k eigenfunctions to efficiently approximate smooth correspondences. The unknown matrix $\boldsymbol{C}$ encodes the functional map T_F, which can be derived by linear constraints such as descriptor and segment preservation together with the operator commutativity (Ovsjanikov et al., 2012).

3.2 Pointwise Map Recovery

Given a functional map T_F, the pointwise map T between two discrete shapes can be reconstructed (Ovsjanikov et al., 2012). Specifically, for any point x on $\mathcal{M}$, its corresponding point $T(x)$ in $\mathcal{N}$ is denoted as $T(x) = \arg\max_y T_F(\delta_x)(y)$, where δ_x is the delta-function at point x on the shape $\mathcal{M}$. As $\langle \delta_x, \phi_i \rangle = \phi_i(x)$, if let the matrices $\boldsymbol{\Phi}_{\mathcal{M}} \in \mathbb{R}^{m \times k}$ and $\boldsymbol{\Phi}_{\mathcal{N}} \in \mathbb{R}^{n \times k}$ respectively denote the first k Laplace-Beltrami eigenfunctions of $\mathcal{M}$ and $\mathcal{N}$, where each column corresponds to an eigenfunction and each row to a point, each row vector of $\boldsymbol{C}\boldsymbol{\Phi}_{\mathcal{M}}^{\top}$ represents transferred Fourier coefficients of a point on $\mathcal{M}$ and has high similarity to a row vector of $\boldsymbol{\Phi}_{\mathcal{N}}^{\top}$ corresponding its matching point on $\mathcal{N}$. By Plancherel's theorem (Penney, 1975), the distances between coefficient vectors of functions can be computed by L^2 differences. Thus, the pointwise map T can be recovered by minimizing the following function:

$$\min_{T} \sum_{i=1}^{m} \left\| \boldsymbol{C}\left(\boldsymbol{\Phi}_{\mathcal{M}}(i)\right)^{\top} - \left(\boldsymbol{\Phi}_{\mathcal{N}}(T(i))\right)^{\top} \right\|_2, \tag{2}$$

where $\boldsymbol{\Phi}_{\mathcal{M}}(i)$ represents the i-th row of matrix $\boldsymbol{\Phi}_{\mathcal{M}}$ and $\boldsymbol{\Phi}_{\mathcal{N}}(T(i))$ corresponds to the mapping point of point i on $\mathcal{N}$. Eq. (2) can be solved by nearest neighbor search via KD tree. In this way, the high-dimensional pointwise maps T can be recovered from the functional spaces, though they may suffer from limited accuracy and continuity.

4 Methodology

This section describes our LORD for pointwise map recovery in functional maps. We propose an effective framework based on local consistency and functional deformation to refine noisy point-to-point correspondences. Given that pointwise maps incorporate both correct correspondences (inliers) and incorrect ones (outliers), our framework involves two steps: (i) distinguishing inliers and outliers from noisy pointwise maps; (ii) refining the point correspondences of outliers based on the reliable mapping of inliers.

4.1 Distinguish Inliers

Given that $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^m$ and $\mathcal{Y} = \{\boldsymbol{y}_i\}_{i=1}^n$ are two sets of vertex spatial coordinates on two discrete manifolds $\mathcal{M}$ and $\mathcal{N}$, respectively, and denoting $T_0 \in \mathbb{R}^m$ as an initial pointwise map that maps each point from $\mathcal{M}$ to a point on $\mathcal{N}$, our first goal is to distinguish the inliers from a noisy point correspondence set $\{\boldsymbol{x}_i, \boldsymbol{y}_{T_0(i)}\}_{i=1}^m$. This process involves two stages: neighborhood consensus and functional deformation.

4.1.1 Local Consistency

By observing the local consistency applicable to various deformations, we devise a simple and efficient mathematical model. The derivation of this model provides a closed-form linear solution that estimates the initial distribution of inliers, described as follows.

Ideal Isometric Formulation Denoting $\mathcal{I}$ as an unknown inlier set and C as a cost function, we have the following objective to find the inliers in the isometric transformation case:

$$\mathcal{I}^* = \arg\min_{\mathcal{I}} C(\mathcal{I}; \mathcal{X}, \mathcal{Y}, T_0, \lambda). \tag{3}$$

Since the distance between two arbitrary points in $\mathcal{M}$ preserves the same as that of their corresponding points in $\mathcal{N}$, the cost function C can be defined as

$$C(\mathcal{I}; \mathcal{X}, \mathcal{Y}, T_0, \lambda) = \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{I}} \left(d(\boldsymbol{x}_i, \boldsymbol{x}_j) - d(\boldsymbol{y}_{T_0(i)}, \boldsymbol{y}_{T_0(j)})\right)^2 + \lambda(|T_0| - |\mathcal{I}|), \tag{4}$$

where d is the geodesic distance, $|\cdot|$ denotes the cardinality of a set and $|T_0| = m$. The first term of Eq. (4) restricts the variance of geodesic distances between any two point pairs, while the second term discourages the outliers with a balance parameter $\lambda > 0$. Ideally, this cost function should be minimized to zero.

General Shape Matching Real applications generally require the acquisition and analysis of nonrigidly deformable objects. Though complex nonrigid deformations produces non-isometric maps, the topology of the local structure remains consistent. In discrete setting, the point distribution in a local region is preserved even after severe deformation. Thus, we have a more general form of Eq. (4):

$$C(\mathcal{I}; \mathcal{X}, \mathcal{Y}, T_0, \lambda) = \sum_{i \in \mathcal{I}} \frac{1}{|N_{\boldsymbol{x}_i}| + |N_{\boldsymbol{y}_{T_0(i)}}|} \left(\sum_{j | \boldsymbol{x}_j \in N_{\boldsymbol{x}_i}} \left(d(\boldsymbol{x}_i, \boldsymbol{x}_j) - d(\boldsymbol{y}_{T_0(i)}, \boldsymbol{y}_{T_0(j)})\right)^2 + \sum_{j | \boldsymbol{y}_{T_0(j)} \in N_{\boldsymbol{y}_{T_0(i)}}} \left(d(\boldsymbol{x}_i, \boldsymbol{x}_j) - d(\boldsymbol{y}_{T_0(i)}, \boldsymbol{y}_{T_0(j)})\right)^2 \right) + \lambda(|T_0| - |\mathcal{I}|), \tag{5}$$

where $N_{\boldsymbol{x}}$ denotes the neighborhood of vertex $\boldsymbol{x}$, which consists of $\boldsymbol{x}$'s adjacent points under triangular meshing. Due to severe deformations, we only assume the distance preservation in a local (neighboring/adjacent) region. Therefore, the distance d in Eq. (5) is:

$$d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \begin{cases} 0, \ \boldsymbol{x}_j \in N_{\boldsymbol{x}_i} \\ 1, \ \boldsymbol{x}_j \notin N_{\boldsymbol{x}_i} \end{cases}, \tag{6}$$

and we also have a similar definition for $d(\boldsymbol{y}_i, \boldsymbol{y}_j)$.

Let a binary vector $\boldsymbol{p}$ represent the correctness of correspondences, where $p_i = 0$ indicates an inlier $(\boldsymbol{x}_i, \boldsymbol{y}_{T_0(i)})$, and $p_i = 1$ indicates an outlier. Substituting Eq. (6) into the cost function Eq. (5), we have

$$\begin{aligned} &C(\boldsymbol{p}; \mathcal{X}, \mathcal{Y}, T_0, \lambda) \\ &= \sum_{i=1}^{m} \frac{p_i}{|N_{\boldsymbol{x}_i}| + |N_{\boldsymbol{y}_{T_0(i)}}|} \Bigg(\sum_{j|\boldsymbol{x}_j \in N_{\boldsymbol{x}_i}} d(\boldsymbol{y}_{T_0(i)}, \boldsymbol{y}_{T_0(j)}) \\ &+ \sum_{j|\boldsymbol{y}_{T_0(j)} \in N_{\boldsymbol{y}_{T_0(i)}}} d(\boldsymbol{x}_i, \boldsymbol{x}_j) \Bigg) + \lambda \big(m - \sum_{i=1}^{m} p_i\big). \end{aligned} \tag{7}$$

In this case, the topological constraints based on local consistency are invariant to translation, rotation, and scale, facilitating the robustness for various deformations.

Optimization To minimize the cost function Eq. (7), we reformulate it by merging the terms related to p_i as follows:

$$C(\boldsymbol{p}; \mathcal{X}, \mathcal{Y}, T_0, \lambda) = \sum_{i=1}^{m} p_i (c_i - \lambda) + \lambda m, \tag{8}$$

where c_i is defined as the disparity score:

$$c_i = 1 - \frac{2n_i}{|N_{\boldsymbol{x}_i}| + |N_{\boldsymbol{y}_{T_0(i)}}|}, \tag{9}$$

and $n_i = \text{count}(j|\boldsymbol{x}_j \in N_{\boldsymbol{x}_i}, \boldsymbol{y}_{T_0(j)} \in N_{\boldsymbol{y}_{T_0(i)}})$ is the number of shared elements in the two neighborhoods. We note that c_i is approximately 0 if $(\boldsymbol{x}_i, \boldsymbol{y}_{T_0(i)})$ is an ideally correct correspondence.

Since a surface usually consists of a vary large number of vertices (*e.g.,* ten thousands), it is necessary to consider multiple neighborhood, rather than just a single one. To better describe the structure of local region, we extend the disparity score c_i into

$$\widetilde{c}_i = \frac{c_i + \sum_{j|\boldsymbol{x}_j \in N_{\boldsymbol{x}_i}} c_j}{1 + |N_{\boldsymbol{x}_i}|}, \tag{10}$$

and the cost function is updated as

$$C(\boldsymbol{p}; \mathcal{X}, \mathcal{Y}, T_0, \lambda) = \sum_{i=1}^{m} p_i (\widetilde{c}_i - \lambda) + \lambda |T_0|. \tag{11}$$

Eq. (11) shows that any correspondence with $\widetilde{c}_i > \lambda$ produces a positive term that increases the cost function, while that with $\widetilde{c}_i < \lambda$ leads to a negative term, thus decreasing the objective. λ is a balance factor that determines the trade-off between the two terms in Eq. (4), *i.e.,* the lower value of λ, the smaller and more reliable the set $\mathcal{I}$ is.

Apparently the consensus score $\{\widetilde{c}_i\}_{i=1}^m$ can be computed in advance. Given triangular meshing and mapping T_0 of two manifolds, the optimal $\boldsymbol{p}$ minimizing Eq. (11) can be obtained by:

$$p_i = \begin{cases} 1, \ \widetilde{c}_i \le \lambda, \\ 0, \ \widetilde{c}_i > \lambda, \end{cases} \quad i = 1, \ldots, m, \tag{12}$$

which would be used as an initial probability $\boldsymbol{P}_0$ for the subsequent probabilistic model, thereby helping to avoid the EM algorithm converging to local optima.

4.1.2 Functional Deformation

The constraint of local consistency alone is insufficient, as it involves only limited contextual information for each vertex, leading to potential false identifications. Additionally, the concise local consistency-based model is sensitive to the preset threshold (*i.e.,* λ). As shown in Fig. 3, distinguishing inliers through local consistency is affected by λ: a high value of λ results in incorrectly identified inliers, whereas a low λ overlooks many inliers. In contrast, the global probability model would exhibit stable and robust. Next, we introduce a functional deformation-based probabilistic model to further distinguish inliers. As described in Myronenko and Song (2010); Eisenberger et al. (2020); Fan et al. (2022), functional deformation aims to align the deformed source set $\mathcal{X}$ with the target set $\mathcal{Y}$ using truncated Laplace-Beltrami eigenbases. Given the deformation as $\mathcal{T}(\cdot)$ and deformation weight matrix as $\boldsymbol{\Psi} \in \mathbb{R}^{k \times 3}$, the deformed coordinate of point $\boldsymbol{x}$ from the source shape is denoted as:

$$\mathcal{T}(\boldsymbol{x}_i) = \boldsymbol{x}_i + \boldsymbol{\Phi}_{\mathcal{M}}(i)\boldsymbol{\Psi}, \tag{13}$$

where the deformed coordinates are expected to closely approximate the target point set. Under the condition that the coordinates $\boldsymbol{x}_i$ and the eigenvectors $\boldsymbol{\Phi}_{\mathcal{M}}(i)$ (i-th row of $\boldsymbol{\Phi}_{\mathcal{M}}$) of each point $\boldsymbol{x}_i$ are known, employing the deformation matrix $\boldsymbol{\Psi}$ at a low dimension k (which is typically much smaller than the total number of points) inherently incorporates smoothness constraints, *i.e.,* points in close proximity are expected to undergo similar deformations. Based on this, we formulate a probabilistic model to estimate functional deformation.

Probabilistic Model Formulation For the putative correspondence set $\mathcal{S} = \{s_i\}_{i=1}^m = \{\boldsymbol{x}_i, \boldsymbol{y}_{T_0(i)}\}_{i=1}^m$ whose mapping relationships are decided by initial pointwise map T_0, our goal is to estimate the underlying deformation, *i.e.,* $\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^3$ between the source point set $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^m$ and

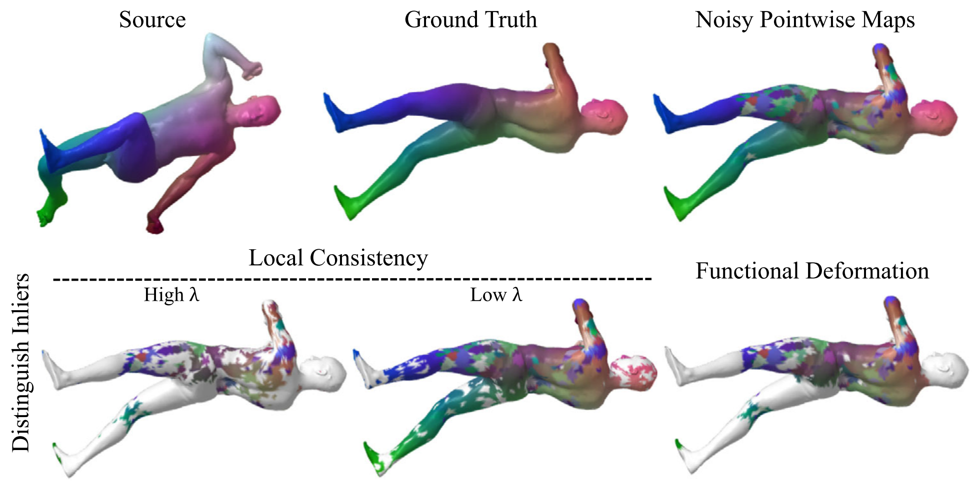

Fig. 3 A visual illustration of inlier (white) identification based on local consistency and functional deformation. Correspondences are visualized via color transfer, where matched points share the same color across the two shapes. The inlier identification guided by functional deformation is robust to the sensitivity of the local consistency regarding λ, resulting in more reliable inlier detection

the mapped target $\mathcal{Y}_{T_0} = \{\boldsymbol{y}_{T_0(i)}\}_{i=1}^{m}$, and identify the correct correspondence (inlier) set $\mathcal{I} \subseteq \mathcal{S}$. In our context, $\boldsymbol{y}_{T_0(i)} = \mathcal{T}(\boldsymbol{x}_i)$ for $\boldsymbol{s}_i \in \mathcal{I}$. According to the conventions in Myronenko and Song (2010); Fan et al. (2022), the source point set $\mathcal{X}$ is assumed to be generalized by a Gaussian Mixture Model (GMM) and $\mathcal{Y}_{T_0}$ is the GMM centroids. We adopt equal isotropic covariance σ^2 and equal membership probabilities. The distribution of outliers is uniform $1/a$ with a denoting the volume of distribution area. Using γ as the weight of outliers, we compute the probability of s_i being inlier as

$$p(s_i) = \frac{1-\gamma}{(2\pi\sigma^2)^{3/2}} e^{-\frac{\|\boldsymbol{y}_{T_0(i)} - \mathcal{T}(\boldsymbol{x}_i)\|^2}{2\sigma^2}} + \gamma\frac{1}{a}. \tag{14}$$

Following the maximum likelihood estimation, the overall objective is determined as $p(\mathcal{S}) = \prod_{i=1}^{m} p(s_i)$.

Expectation-Maximization Based Solution Using the effective EM algorithm, we derive the optimal solution through the iteration of an expectation (E) step and a maximization (M) step. The E-step involves computing a posterior probabilities for the mixture components, assuming $p_i = p(s_i)$, as:

$$p_i = \frac{(1-\gamma)e^{-\frac{\|\boldsymbol{y}_{T_0(i)} - \mathcal{T}(\boldsymbol{x}_i)\|^2}{2\sigma^2}}}{(1-\gamma)e^{-\frac{\|\boldsymbol{y}_{T_0(i)} - \mathcal{T}(\boldsymbol{x}_i)\|^2}{2\sigma^2}} + \gamma\frac{(2\pi\sigma^2)^{3/2}}{a}}. \tag{15}$$

The M-step updates parameters by minimizing the expectation of the complete negative log likelihood function as

$$\mathcal{Q}\left(\theta, \sigma^2\right) = \sum_{i=1}^{l} p_i \frac{\left\|\boldsymbol{y}_{T_0(i)} - \mathcal{T}(\boldsymbol{x}_i)\right\|^2}{2\sigma^2} + \frac{3}{2}\log\sigma^2 \sum_{i}^{l} p_i. \tag{16}$$

Applying Eq. (13) to Eq. (16), we obtain a closed-form solution for the deformation matrix $\boldsymbol{\Psi}$ by

$$\boldsymbol{P}\boldsymbol{\Phi}_{\mathcal{M}}\boldsymbol{\Psi} = \boldsymbol{P}\boldsymbol{Y} - \boldsymbol{P}\boldsymbol{X}, \tag{17}$$

where $\boldsymbol{P}$ is a diagonal matrix consisting of posterior probability distribution, *i.e.*, $\boldsymbol{P} = diag([p_i])$, and $\boldsymbol{X} \in \mathbb{R}^{m\times 3}$ and $\boldsymbol{Y} \in \mathbb{R}^{m\times 3}$ stack $\boldsymbol{x}_i$ and $\boldsymbol{y}_{T_0(i)}$ of correspondences $\{s_i\}_{i=1}^{m}$. Based on linear algebra, we have

$$\sigma^2 = \frac{tr\left((\boldsymbol{Y} - \boldsymbol{\Gamma})^\top \boldsymbol{P}(\boldsymbol{Y} - \boldsymbol{\Gamma})\right)}{3tr(\boldsymbol{P})}, \tag{18}$$

$$\gamma = 1 - \frac{tr(\boldsymbol{P})}{N}, \tag{19}$$

where $\boldsymbol{\Gamma} = \boldsymbol{X} + \boldsymbol{\Phi}_{\mathcal{M}}\boldsymbol{\Psi}$. Hence, the complete EM algorithm is completed by conducting a generalized probabilistic model.

Hence, with ϵ as a predefined constant threshold and empirically set to 0.95, the optimal inlier set $\mathcal{I}^*$ is

$$\mathcal{I}^* = \{i | p_i > \epsilon, i = 1, \cdots, m\}. \tag{20}$$

4.2 Correspondence Refinement

After obtaining the inlier correspondence set and the functional deformation model, the next step is to refine the correspondences for outliers. To this end, we propose a method that incorporates both local and global topological constraints to find appropriate corresponding points on the target shape for the outliers on the source shape.

Specifically, we divide the vertex set $\mathcal{X}$ into two parts: $\mathcal{X}_{\mathcal{I}}$ indicates inliers with correct correspondences, and $\mathcal{X}_{\mathcal{O}}$ denotes the outliers with mismatches. We also have similar sets $\mathcal{Y}_{\mathcal{I}}$ and $\mathcal{Y}_{\mathcal{O}}$ for $\mathcal{Y}$. Next, we show how to choose a corresponding vertex from $\mathcal{Y}_{\mathcal{O}}$ for each point in $\mathcal{X}_{\mathcal{O}}$ based on reliable maps $T_{\mathcal{I}} = (\mathcal{X}_{\mathcal{I}}, \mathcal{Y}_{\mathcal{I}})$ and estimated deformation $\mathcal{T}(\cdot)$ while introducing geometric constraints based on LLE.

Firstly, for each point $\mathbf{x}_i^{\mathcal{O}}$ of $\mathcal{X}_{\mathcal{O}}$, we search its K_1 nearest neighbors from $\mathcal{X}_{\mathcal{I}}$ using the L_2 distance of their coordinates on manifold $\mathcal{M}$, obtaining a neighborhood set $N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1}$:

$$N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1} = K_1\text{-NNsearch}(\mathbf{x}_i^{\mathcal{O}}, \mathcal{X}^{\mathcal{I}}), \tag{21}$$

where $\mathbf{x}_i^{\mathcal{O}}$ is the searching center, K_1 is the number of neighbors, and $\mathcal{X}^{\mathcal{I}}$ is the searching space. In this step, $n_{\mathcal{O}}$ neighborhood sets will be generated, where $n_{\mathcal{O}} = |\mathcal{X}_{\mathcal{O}}|$ is the number of points in $\mathcal{X}_{\mathcal{O}}$.

Secondly, to derive a weight matrix $\mathbf{W} \in \mathbb{R}^{n_{\mathcal{O}} \times K_1}$, we minimize the reconstruction errors measured by the following cost function:

$$\mathcal{E}(\mathbf{W}) = \sum_{i=1}^{n_{\mathcal{O}}} \left\| \mathbf{x}_i^{\mathcal{O}} - \sum_{j=1}^{K_1} \mathbf{W}_{ij} \mathbf{x}_{i,j} \right\|^2, \text{ s.t. } \forall i, \sum_{j=1}^{K_1} \mathbf{W}_{ij} = 1, \tag{22}$$

where $\forall i, \mathbf{x}_{i,j} \in N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1}$, $j = 1, \ldots, K_1$, and the topology of vertex distribution is preserved in the weight matrix $\mathbf{W}$ via the neighboring point relationship. We can efficiently solve Eq. (22) for $\mathbf{W}_{ij}$ using the least squares.

Thirdly, we seek the corresponding vertex $\mathbf{y}_{T_{\mathcal{O}}(i)}^{\mathcal{O}}$ from $\mathcal{Y}_{\mathcal{O}}$ for $\mathbf{x}_i^{\mathcal{O}}$ from $\mathcal{X}_{\mathcal{O}}$, to obtain the pointwise map $T_{\mathcal{O}}$ for outliers. We solve this in an embedded manifold representation using spectral similarity. Specifically, we obtain K_1 corresponding points $T_{\mathcal{I}}(N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1})$ of neighborhood $N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1}$ using the pointwise map $T_{\mathcal{I}}$. Then, we find the K_2 nearest neighbors between the column vectors of $\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}})^\top$ and $(\Phi_{\mathcal{N}}^{\mathcal{O}})^\top$ as the spectral neighborhoods, where $\Phi_{\mathcal{M}}^{\mathcal{O}}/\Phi_{\mathcal{N}}^{\mathcal{O}} \in \mathbb{R}^{n_{\mathcal{O}} \times k}$ is the matrix consisting of the row vectors of $\Phi_{\mathcal{M}}/\Phi_{\mathcal{N}}$ corresponding to outliers $\mathcal{X}_{\mathcal{O}}/\mathcal{Y}_{\mathcal{O}}$, respectively.

Combined with the estimated functional deformation, the correspondence $\mathbf{y}_{T_{\mathcal{O}}(i)}^{\mathcal{O}}$ for the vertex $\mathbf{x}_i^{\mathcal{O}}$ can be chosen as the vertex with the minimal reconstruction error from the spectral neighborhood $\mathcal{N}_{\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}}(i))^\top}^{K_2}$:

$$T_{\mathcal{O}}(i) = \underset{j|\Phi_{\mathcal{N}}(j) \in \mathcal{N}_{\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}}(i))^\top}^{K_2}}{\arg\min} \left\| \mathbf{y}_j - \frac{1}{2}\left(\sum_{k=1}^{K_1} \mathbf{W}_{ik} \mathbf{y}_{i,k} + \mathcal{T}(\boldsymbol{x}_i) \right) \right\|^2, \tag{23}$$

where the spectral neighborhood

$$\mathcal{N}_{\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}}(i))^\top}^{K_2} = K_2\text{-NNsearch}(\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}}(i))^\top, (\Phi_{\mathcal{N}}^{\mathcal{O}})^\top), \tag{24}$$

Algorithm 1: The overall algorithm flow

Input: A pair of discrete manifolds $\mathcal{M}$ and $\mathcal{N}$ with bases $\boldsymbol{\Phi}_{\mathcal{M}}$ and $\boldsymbol{\Phi}_{\mathcal{N}}$ and a functional map $\mathbf{C}$
Parameter: λ
Output: Pointwise map T
1: Obtain T_0 by selecting NN using Eq. (2);
2: Calculate disparity scores $\{\widetilde{c}_i\}_{i=1}^m$ by Eq. (10);
3: Determine initial probability $\boldsymbol{P}_0$ by Eq. (12);
4: **while** $\mathcal{Q}$ not converge **do**
5: *M-step:*
6: Update $\boldsymbol{\Psi}$ and $\boldsymbol{\Gamma}$ by (17) and (13);
7: Update σ^2 and γ by (18) and (19);
8: *E-step*:
9: Update $\boldsymbol{P}$ by (15);
10: **end while**
11: Obtain optimal inlier set $\mathcal{I}^*$ by (20);
12: Construct neighborhoods $\{N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1}\}_{i=1}^{n_{\mathcal{O}}}$ by Eq. (21);
13: Compute matrix $\mathbf{W}$ by minimizing Eq. (22);
14: Obtain outlier map $T_{\mathcal{O}}$ by Eqs. (23) and (24);
15: Obtain T by combining $T_{\mathcal{O}}$ and $T_{\mathcal{I}}$ from $\mathcal{I}^*$;

and $\forall i$, $\mathbf{x}_i^{\mathcal{O}} \in \mathcal{X}_{\mathcal{O}}$, $\mathbf{y}_{i,k} \in T_{\mathcal{I}}(N_{\mathbf{x}_i^{\mathcal{O}}}^{K_1})$, $k = 1, \ldots, K_1$. In Eq. (24), $\mathbf{C}(\Phi_{\mathcal{M}}^{\mathcal{O}}(i))^\top$ is the searching center, K_2 is the number of neighbors, and $(\Phi_{\mathcal{N}}^{\mathcal{O}})^\top$ is the searching space. Finally, the pointwise maps $T_{\mathcal{I}}$ for inliers and $T_{\mathcal{O}}$ derived from Eq. (23) form a full pointwise map T. The overall algorithm flow is described in Algorithm 1.

The functional map framework commonly iterates between the estimation of functional map matrix $\mathbf{C}$ and the recovery of pointwise maps T. Taking the representative MWP (Hu et al., 2021) as an example, when our LORD is applied to its iterative process, more accurate point-to-point correspondences are achieved. As illustrated in Fig. 4, in each iteration, the two-step process of LORD (distinguish inliers and outlier refinement) results in more precise pointwise maps. With an increasing number of iterations, LORD attains matching results that closely approximate the ground truth (GT). Naturally, our LORD can also be used in other functional mapping methods because of its generalizability, and the related experiments are shown in Sect. 5.5.1.

4.3 Parameter Setting

Our method introduces three hyperparameters, i.e., λ, K_1, K_2. We explain how to set these parameters below.

Parameter λ is the crucial threshold for the initial probability distribution of the inlier set $\mathcal{I}$. To find an appropriate value, we selected 300 shape pairs from the FAUST dataset (Bogo et al., 2014) as the test set and used the complete LORD for pointwise map recovery of MWP (Hu et al., 2021). As shown in Fig. 5, λ can be set to 0.4 to achieve the lowest matching error.

Parameter K_1 determines the number of nearest neighbors in Eq. (21), which is used to construct local manifold

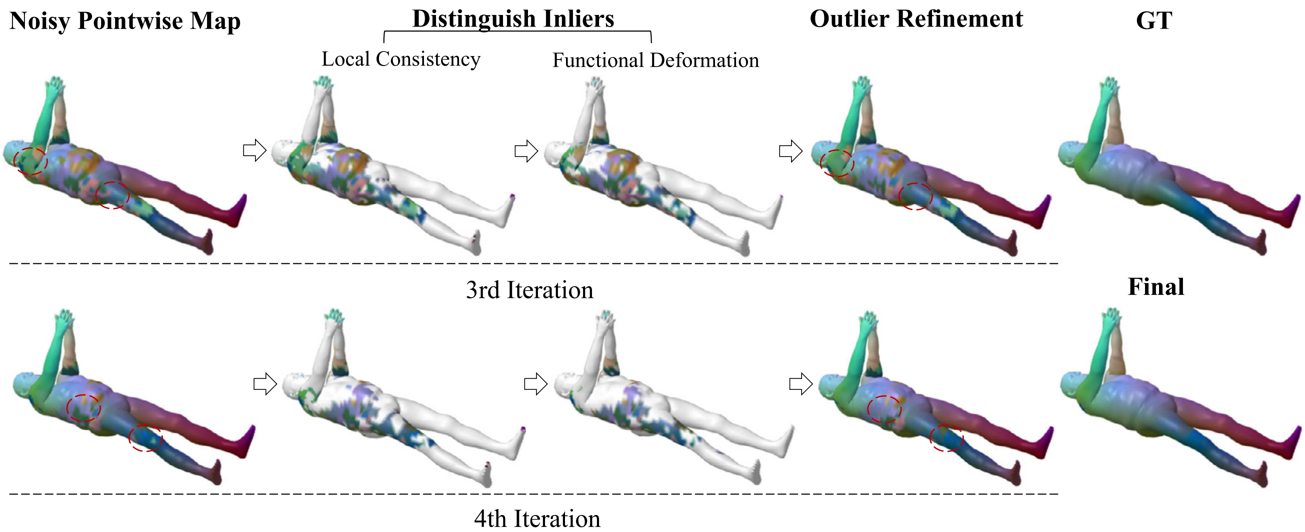

Fig. 4 A visual illustration demonstrates the process of our LORD used in two iterations of MWP (Hu et al., 2021), where the pointwise map recovery (*i.e.,* outlier refinement after distinguishing inliers) yields noticeable improvements

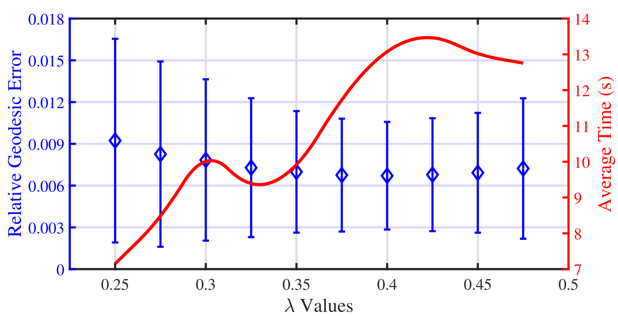

Fig. 5 Parameter analysis of λ. Performance of LORD under different values of λ is evaluated using the mean and standard deviation of errors (Blue), with the average time displayed on the right (Red)

representations in Eq. (22). Intuitively, a higher value of K_1 imposes stronger local geometric constraints but increases computational time. Considering that a larger set $\mathcal{I}$ can support more nearest neighbors for LLE, we empirically set an adaptive value for K_1 *w.r.t.* the cardinality of $\mathcal{I}$, *e.g.,* $K_1 = \lceil |\mathcal{I}|/100 \rceil$, where $\lceil \cdot \rceil$ denotes rounding up. Similarly, as the number of nearest neighbors searched in the spectral domain in Eq. (24), K_2 should also be proportional to $|\mathcal{I}|$, *e.g.,* $K_2 = \lceil |\mathcal{I}|/1000 \rceil$.

4.4 Computational Complexity

Our LORD involves two main steps: inlier identification and correspondence refinement.

4.4.1 Inlier Identification

The computational cost in this step primarily comes from computing the disparity score $\widetilde{c}_i$ (Eq. (10)) and the posterior probability p_i (Eq. (15)). When shape meshing provides adjacencies, both calculations have a computational complexity of $O(N)$, where N is the number of points on the manifold $\mathcal{M}$ (*i.e., m* in Sect.. 4).

4.4.2 Correspondence Refinement

The computational cost in this step depends on two K-nearest neighbor searches using a K-D tree (*i.e.,* Eq. (21) and Eq. (24)), with the complexities of $O((K_1 + N_{\mathcal{O}}) \log N_{\mathcal{O}})$ and $O((K_2 + N_{\mathcal{O}}) \log N_{\mathcal{O}})$, where $N_{\mathcal{O}}$ is the number of outliers $\mathcal{X}_{\mathcal{O}}$ on the manifold $\mathcal{M}$. Since $K_1 \ll N_{\mathcal{O}}$ and $K_2 \ll N_{\mathcal{O}}$, the time complexity of this step can be simplified to $O(N_{\mathcal{O}} \log N_{\mathcal{O}})$, and the space complexity is easily identified as $O(N_{\mathcal{O}})$.

Given that $N_{\mathcal{O}}$ and N are of the same order of magnitude, the proposed LORD exhibits linearithmic complexity for both time and space *w.r.t.* the number of points N on the 3-dimensional manifold. This ensures the high efficiency of our LORD.

5 Experiment

In this section, we apply our LORD to challenging shapes from several public datasets and compare its performance with both classical and state-of-the-art approaches.

5.1 Implement Details

5.1.1 Datasets

Six public datasets are used in our evaluation experiments:

- **FAUST** (Bogo et al., 2014) contains 100 shapes representing 10 poses of 10 different human subjects, exhibiting significant variations across different subjects. Each shape comprises 6890 element vertices, making it suitable for computationally demanding methods. For quantitative evaluation, we randomly selected 300 shape pairs, encompassing both isometric and non-isometric deformations.
- **TOSCA** (Bronstein et al., 2008) provides 76 shapes divided into 8 distinct categories (ranging from human to animal), with almost every shape containing 10k vertices. Our experimental evaluation involves all isometric shape pairs, totaling 414 pairs.
- **SCAPE** (Anguelov et al., 2005) contains 71 registered meshes representing different poses of the same human subject. Each mesh consists of 12500 vertices. We randomly selected 200 matching pairs for quantitative evaluations.
- **TOPKIDS** (Lähner et al., 2016) consists of 26 non-intersecting manifold shapes generated by merging self-intersecting parts of the shapes from KIDS dataset (Rodola et al., 2014), making it highly challenging. We randomly selected 200 matching pairs from low-resolution shapes (each mesh contains approximately 10k vertices) for quantitative evaluations.
- **SHREC'19** (Dyke et al., 2019) contains 44 human shapes undergoing various controlled deformations (articulating, bending, stretching, and topological changes), and each shape has about 10400 vertices. This dataset provides 430 pointwise maps corresponding to 430 shape pairs as the ground-truth.
- **DT4DM** (Li et al., 2021) spans 10 different animal or humanoid categories with dense 4D annotation, each containing 31 objects. We randomly selected 60 pairs of shapes from each category, resulting in a total of 600 pairs for evaluation.

5.1.2 Evaluation

We use the Princeton benchmark protocol (Kim et al., 2011) to evaluate the accuracy of shape matching. Specifically, given a ground-truth correspondence $(\mathbf{x}, \mathbf{y}^*)$ where $\mathbf{x} \in \mathcal{X}$ and $\mathbf{y}^* \in \mathcal{Y}$, the error for the obtained correspondence $(\mathbf{x}, \mathbf{y})$ is calculated by relative geodesic distance between $\mathbf{y}$ and $\mathbf{y}^*$, normalized by the diameter of $\mathcal{Y}$: $\epsilon(x) = \frac{d_{\text{geo}}(y, y^*)}{\sqrt{\text{area}(\mathcal{Y})}}$. We employ the Correspondence Quality Characteristics (CQC) curves (Kim et al., 2011), which depict the percentages of matches that have geodesic errors no greater than r, and the average geodesic error across all vertices on the shape $\mathcal{M}$, to quantify the correspondence quality.

5.1.3 Competitors

The competitor methods include ICP (Ovsjanikov et al., 2012), BCICP (Ren et al., 2018), ZoomOut (Melzi et al., 2019b), DiscreteOp (Ren et al., 2021), SmoothShells (Eisenberger et al., 2020), MWP (Hu et al., 2021), Sinkhorn (Pai et al., 2021), GCPD (Fan et al., 2022), PC-GAU (Colombo et al., 2023), UNS-FMNet (Cao et al., 2023), and LOPR (Xia et al., 2024). ICP is an initial iterative method based on nearest neighbor search. BCICP extends ICP by incorporating orientation preservation and bijectivity constraints. ZoomOut introduces iterative upsampling in the spectral domain. DiscreteOp designs a discrete solver for pointwise map recovery. SmoothShells fuses extrinsic shape registration with functional maps using an intrinsic-extrinsic product space. MWP introduces multi-scale spectral manifold wavelet consensus into functional maps. Sinkhorn proposes a spectral registration technique to address pointwise map recovery, which is also the focus of our LORD. GCPD extends CPD to the non-Euclidean domain and employs externally-deformation-based probabilistic modeling. PC-GAU uses Principal Components of a Dictionary basis to describe the functional spaces on surfaces. UNS-FMNet is a recent functional map-based deep learning method with state-of-the-art performance. LOPR is a previous version of LORD, based on local constraints to recover pointwise maps.

5.1.4 Settings

The matching results in the SHOT (Tombari et al., 2010) descriptor space are used as initialization for ICP, BCICP, DiscreteOp, and MWP. GCPD uses the results of MWP as its initialization, and similarly to PC-GAU. Sinkhorn and our LOPR are used to recover pointwise maps of MWP with 5 discrete filters and 4 iterations. For all competitors, we use the settings and codes provided online by their authors. In particular, the number of eigenfunctions is uniformly set to 500, which for ZoomOut is the maximal dimension of its upsampling iterations. As suggested by its authors, GCPD is initialized by MWP with 200 eigenfunctions, and the subsequent smoothing model is still based on a 500-dimensional eigenfunction space. All experiments are conducted on a PC with Intel(R) Core i9-9920X CPU at 3.50GHz. Additionally, K-nearest neighbor search and UNS-FMNet (Cao et al., 2023) are accelerated by GPU.

5.2 Results Analysis

The quantitative evaluations on six public datasets include CQC curves and average errors, which are shown in Fig. 6 and Table 1, respectively. Recent State-Of-The-Art (SOTA) methodologies, including MWP (Hu et al., 2021), SinkHorn (Pai et al., 2021), GCPD (Fan et al., 2022), LOPR (Xia et al.,

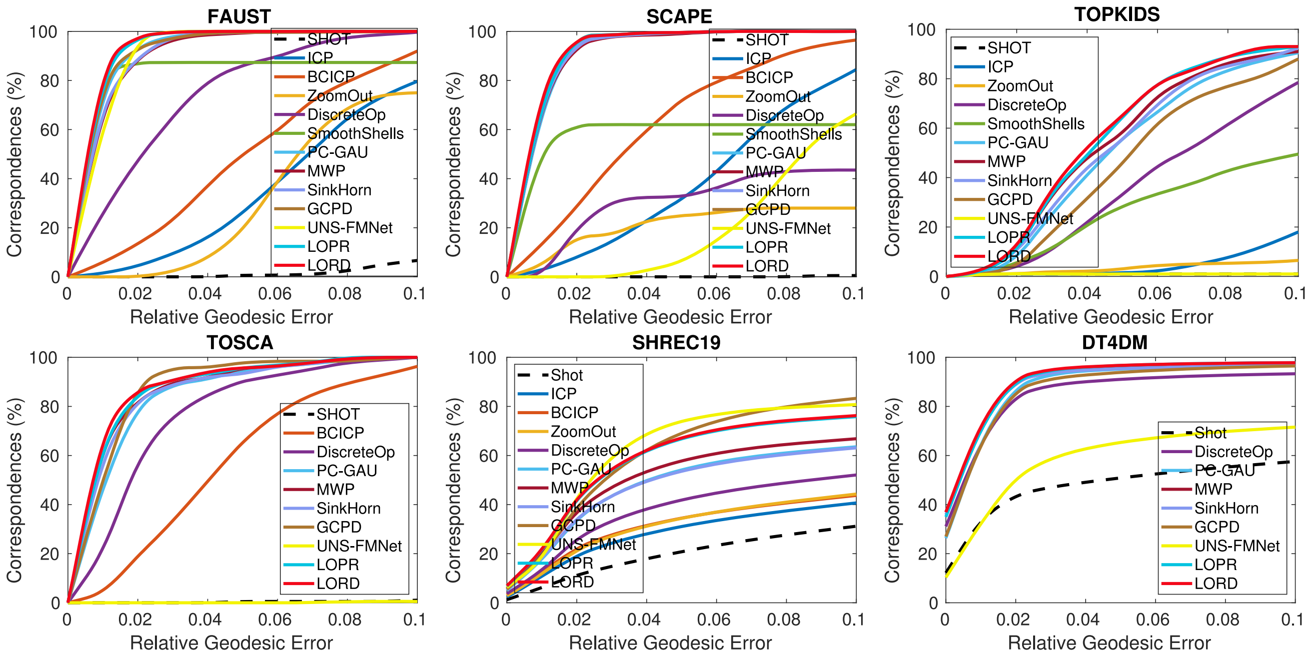

Fig. 6 Evaluations of our LORD with comparative other methods on FAUST (Bogo et al., 2014), SCAPE (Anguelov et al., 2005), TOPKIDS (Lähner et al., 2016), TOSCA (Bronstein et al., 2008), SHREC19 (Dyke et al., 2019), and DT4DM (Li et al., 2021) datasets with CQCs

Table 1 Average relative geodesic errors of our LORD with state-of-the-art comparators on six public datasets, where the **bold** indicates the best and the <u>underline</u> indicates the second

Methods	FAUST	SCAPE	TOPKIDS	TOSCA	SHREC19	DT4DM
DiscreteOp	0.0263	0.1674	0.0770	0.0286	0.1955	0.0334
MWP	0.0100	0.0096	0.0502	0.0171	0.1430	0.0198
SinkHorn	0.0089	0.0096	0.0515	0.0175	0.1530	0.0205
PC-GAU	0.0082	0.0098	0.0542	0.0188	0.1546	0.0225
GCPD	0.0084	**0.0087**	0.0709	0.0159	**0.0789**	0.0208
UNS-FMNet	0.0098	0.0894	0.3163	0.2011	0.1144	0.1306
LOPR	<u>0.0070</u>	0.0091	<u>0.0466</u>	<u>0.0157</u>	0.1146	<u>0.0173</u>
LORD	**0.0067**	**0.0087**	**0.0455**	**0.0154**	<u>0.1131</u>	**0.0170**

2024), and PC-GAU (Colombo et al., 2023), exhibit significant advantages over their predecessors on these datasets. UNS-FMNet (Cao et al., 2023) is a recent state-of-the-art learning-based method, but its model depends on the training data and is competitive on specific datasets (e.g., FAUST and SHREC19), yet fails against other datasets. However, our LORD consistently achieves lower average errors across the majority of datasets, as illustrated in Table 1. This is because our LORD not only embeds external geometric constraints into the functional space, but also exploits both local and global topological information, *i.e.*, local consistency and global functional deformation, to more accurately align shape pairs.

To further examine robustness to topological noise, we evaluate LORD on the challenging TOPKIDS dataset, where shapes undergo structural changes that alter local mesh connectivity (*e.g.*, arms moving from touching the torso to extended positions). Despite such topological inconsistencies, LORD performs well due to its hybrid strategy: local adjacency is used for initial inlier detection, while a global functional deformation model refines correspondences by capturing consistent shape variations without relying on strict topological preservation. As shown in Fig. 6 and Fig. 7, our method consistently outperforms baselines even under significant topological noise, confirming its resilience to topological inconsistencies and practical applicability to real-world non-rigid matching scenarios.

A qualitative comparison with SOTA function mapping methods is displayed in Fig. 7, where each row shows a representative shape pair from each dataset. Compared to the competitors, the matching results of our LORD are more continuous and smooth in color and closer to GT. This comparison visually demonstrates the superior ability of our method in solving challenging matching pairs.

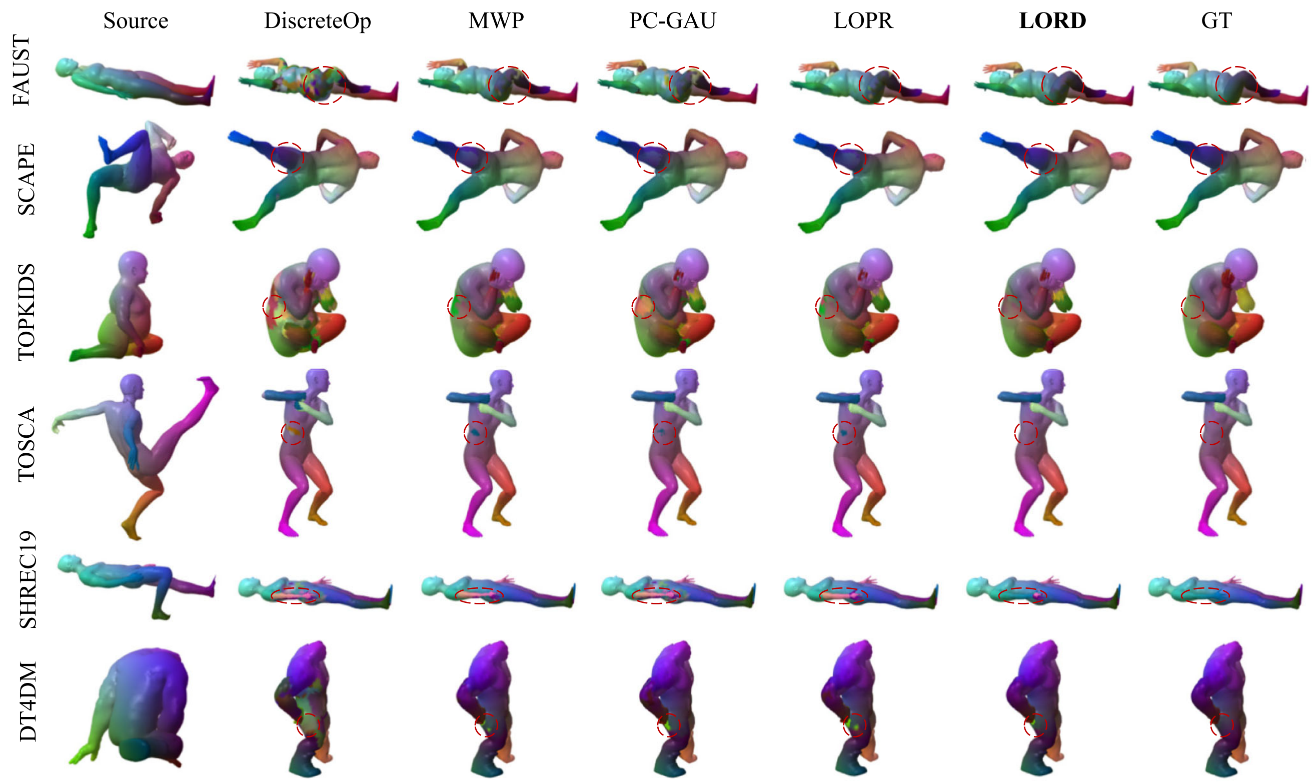

Fig. 7 Qualitative demonstration using color transfer. In each row of results, the first and last shapes represent the source and target shapes matched with the ground truth, respectively. For the five shapes in the middle, from left to right, the results of DiscreteOp (Ren et al., 2021), MWP (Hu et al., 2021), PC-GAU (Colombo et al., 2023), LOPR (Xia et al., 2024), and LORD are shown. From top to bottom, the images in each row are from FAUST (Bogo et al., 2014), SCAPE (Anguelov et al., 2005), TOPKIDS (Lähner et al., 2016), TOSCA (Bronstein et al., 2008), SHREC'19 (Dyke et al., 2019), and DT4DM (Li et al., 2021) datasets, respectively

For the pointwise map recovery problem, five methods are used in MWP framework for comparison, including classic Nearest Neighbors (NN) (Ovsjanikov et al., 2012), recent SinkHorn (Pai et al., 2021), GCPD (Fan et al., 2022), LOPR (Xia et al., 2024), and our LORD. The qualitative results are displayed in Fig. 8. Here, the iterative process of MWP is uniformly set to 5 iterations. Notably, our LORD recovers more accurate pointwise correspondences compared to others, due to their under-utilization of geometric constraints.

Furthermore, the average runtimes of our method, alongside several representative techniques, are reported in Table 2. The runtime of our LORD is deemed acceptable across various shape resolutions (different number of vertices).

5.3 Map Smoothness

While geodesic errors reflect global accuracy, it may overlook local smoothness that affect practical tasks such as texture or mesh transfer. As shown in Fig. 9, LORD yields smoother pointwise correspondences, as illustrated by the vertex transfer results. To quantitatively assess the local smoothness, we adopt *conformal distortion* as the evaluation metric, which measures how much each triangle is distorted under the induced deformation, following Eq. (3) in Hormann and Greiner (2000). We compute the average conformal distortion across all triangles in the deformed mesh. Table 3 reports the comparison among several methods. LORD achieves the lowest average distortion, highlighting its superior ability to preserve local geometric consistency over baselines.

5.4 Partial Matching

The geometric constraints embedded in LORD are based on local consistency and functional deformations, showcasing broad applicability, even in partial matching scenarios. We evaluate on the cuts and holes datasets provided by Partial Functional Correspondences (Rodolà et al., 2017), which includes 456 cut shapes and 152 hole-drilled shapes across different classes of TOSCA. We compare our LORD with representative functional map methods for partial-to-full matching. Qualitative instances comparing our LORD with the SOTA Sinkhorn and LOPR are presented in Fig. 10, demonstrating that our LORD achieves more accurate and

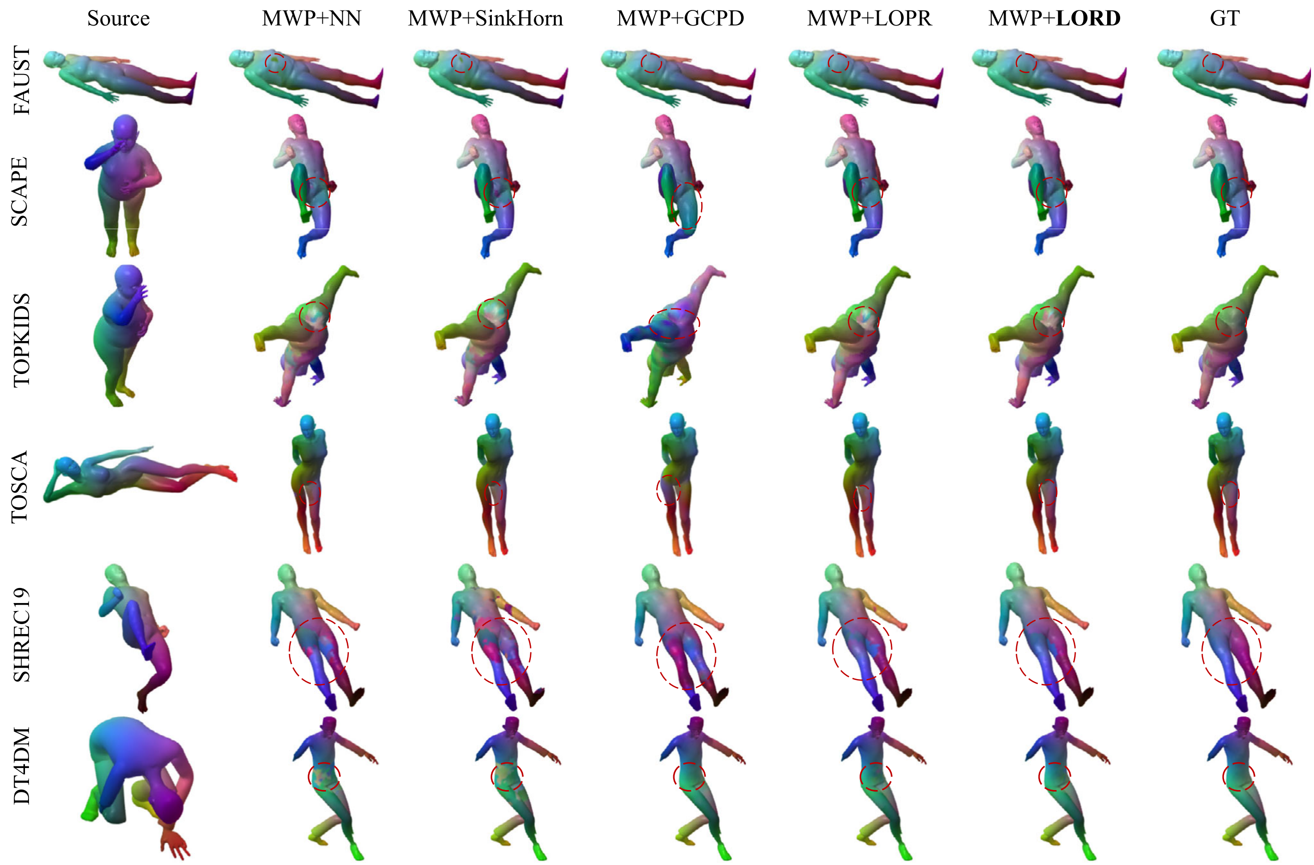

Fig. 8 Qualitative comparisons for NN (Ovsjanikov et al., 2012), Sinkhorn (Pai et al., 2021), GCPD (Fan et al., 2022), LOPR (Xia et al., 2024), and our LORD on MWP (Hu et al., 2021) using color transfer. From top to bottom, the images in each row are from FAUST (Bogo et al., 2014), SCAPE (Anguelov et al., 2005), TOPKIDS (Lähner et al., 2016), TOSCA (Bronstein et al., 2008), SHREC'19 (Dyke et al., 2019), and DT4DM (Li et al., 2021) datasets, respectively

Table 2 Average runtime (in seconds) demonstration of our LORD and state-of-the-arts on different number of vertices. We present the results on FAUST (Bogo et al., 2014), SCAPE (Anguelov et al., 2005), TOPKIDS (Lähner et al., 2016), SHREC'19 (Dyke et al., 2019), DT4DM (Li et al., 2021) datasets, and two models (Wolf and Michael) from TOSCA (Bronstein et al., 2008)

Methods #Vertices	FAUST 6890	SCAPE 12500	TOPKIDS 10399	Wolf 4344	Michael 10000	SHREC19 5202	DT4DM 7996
DiscreteOp	161.88	158.16	76.359	17.540	41.420	17.050	27.043
MWP	1.8358	4.3591	3.6495	0.8974	3.1439	1.7166	2.7447
PC-GAU	21.760	83.052	128.34	8.3798	48.573	13.656	32.425
GCPD	9.3658	20.146	14.184	2.8385	12.615	4.4362	8.6500
SinkHorn	24.072	101.70	63.872	10.010	64.415	31.719	65.488
UNS-FMNet	64.439	87.970	14.397	5.8461	13.587	7.8050	32.850
LOPR	3.9523	13.357	9.4687	1.4258	7.1651	3.8335	5.9351
LORD	15.877	31.409	36.690	2.2712	10.851	16.342	16.433

continuous pointwise maps. From quantitative CQC curves and runtime results illustrated in Fig. 11, it is evident that our method achieves competitive accuracy improvements within a moderate time cost compared to other methods.

5.5 Further Analysis

5.5.1 Generalizability Verification

We provide a generalizability verification for our proposed LORD, with the experimental results presented on the left of Fig. 12. Using the challenging TOPKIDS (Lähner et al., 2016) dataset, we combine LORD with four functional map

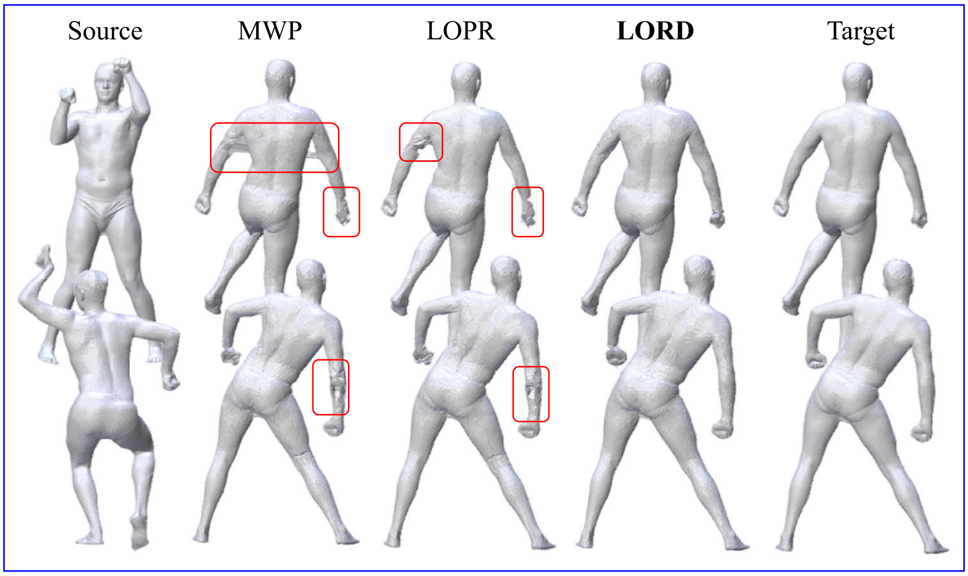

Fig. 9 Visualized comparison based on vertex transfer. After mapping points from the source to the target using different shape matching methods, we visualize the mesh structure of the mapped shape by retaining the original mesh connectivity on the source. Compared with other state-of-the-art results, LORD produces markedly smoother correspondences

Table 3 Average conformal distortion comparisons.

SCAPE	ICP	MWP	GCPD	LOPR	**LORD**
Avg. Distortion	8.08	4.30	3.38	3.38	**3.14**

methods: ZoomOut (Melzi et al., 2019b), DiscreteOp (Ren et al., 2021), PC-GAU (Colombo et al., 2023), and MWP (Hu et al., 2021). These methods represent different improvements on the original functional map framework (Ovsjanikov et al., 2012). ZoomOut changes the iterative strategy by utilizing functional upsampling, DiscreteOp proposes a discrete solver for functional map matrix, PC-GAU designs a new functional basis instead of LB operator, and MWP incorporates wavelet filters to solve functional map matrix. When combined with LORD, all these methods achieve higher matching accuracy. This demonstrates that our LORD can be applied to various types of functional maps, effectively addressing the shortcomings in pointwise map recovery.

5.5.2 Ablation Study

To specifically analyze the effect of technical components in our LORD, we conduct an ablation study, with the results of which presented on the right of Fig. 12. **Locality preserving-based Initialization** (L.I.) constructs initial inlier distributions based on local consistency, avoiding local optima in subsequent deformation estimation. Without LI, a uniform distribution is used for initialization. **LLE-based refinement** (LLE) finds suitable correspondences for outliers via local spectral reconstruction. Without LLE, the nearest neighbor search for outliers relies only on functional deformations. **Deformation-based Refinement** (D.R.) refines outliers using functional deformations; without DR, refinements depend solely on LLE. **Functional Deformation** (F.D.) includes a general probabilistic model and outlier refinement based on functional deformation; without FD, it reverts to the original LOPR. Quantitative results in Fig. 12 show that omitting any component reduces the matching accuracy of LORD, especially without LLE.

6 Limitations and Conclusion

In this work, we proposed Locality Optimization Refinement with Deformation (LORD), a novel strategy for nonrigid shape matching. LORD effectively integrates both local and global constraints in the spectral domain and introduces a novel two-stage pipeline to refine noisy pointwise maps.

However, the proposed LORD also presents certain limitations. Due to the EM optimization, LORD incurs a significant increase in runtime. Future work could explore more efficient optimizations to accelerate convergence. Moreover, the current formulation assumes near-isometric deformations, which may restrict performance in highly non-isometric cases such as matching between a tiger and a pig. This calls for further investigation into more flexible and semantic-aware constraints.

Despite these limitations, the comprehensive experiments conducted on six public datasets demonstrate the effectiveness of LORD in handling diverse shape deformations, including the challenging partial matching settings in the cuts and holes benchmarks. Ablation studies further verify the contributions of each component within the proposed pipeline. Overall, LORD presents a principled and robust framework for shape correspondence, contributing useful insights and tools toward advancing non-rigid shape analysis.

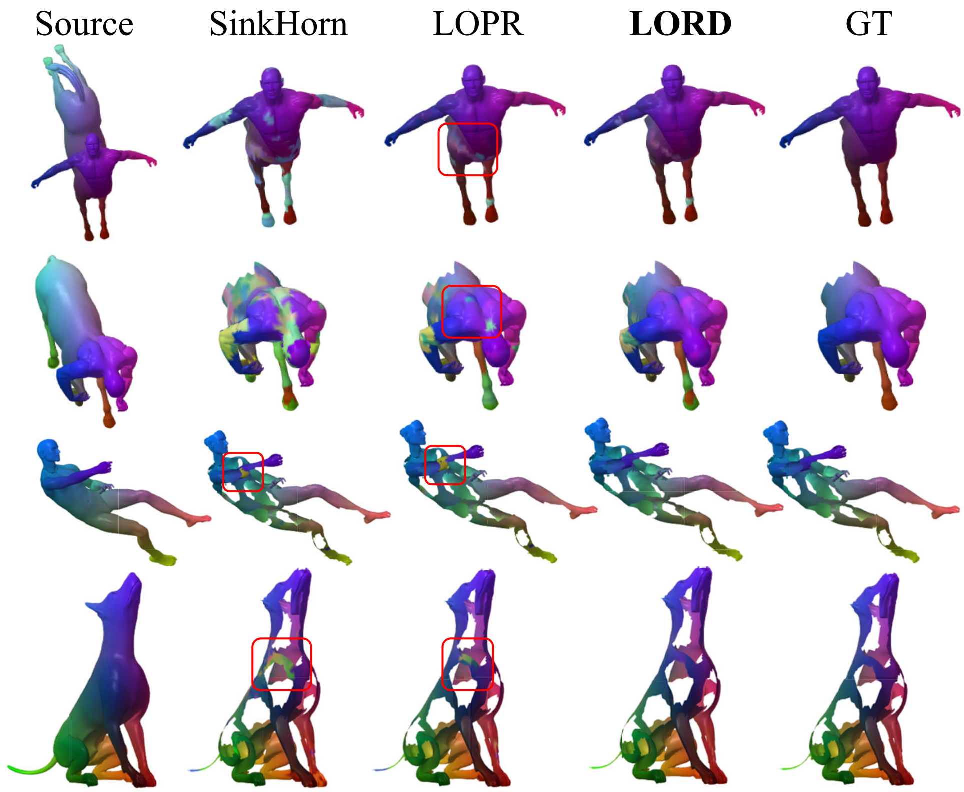

Fig. 10 Qualitative comparisons of our LORD, LOPR, and Sinkhorn for partial shape matching via color transfer. The first two rows correspond to cuts, and the latter two to holes. In each group, the first and last shapes are the source and target shapes with ground-truth correspondences. The second shape shows the result of Sinkhorn, the third of LOPR, and the fourth of our LORD

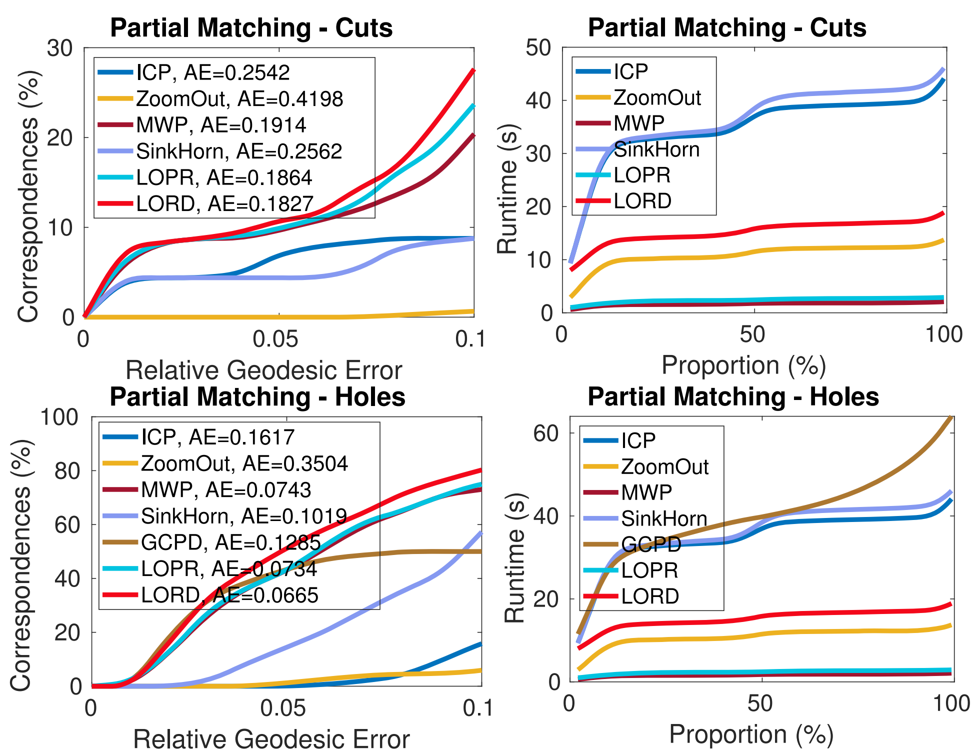

Fig. 11 Evaluations of our LORD method compared to other methods on partial shape matching tasks using CQCs, with the TOSCA cuts and holes as the assessment datasets

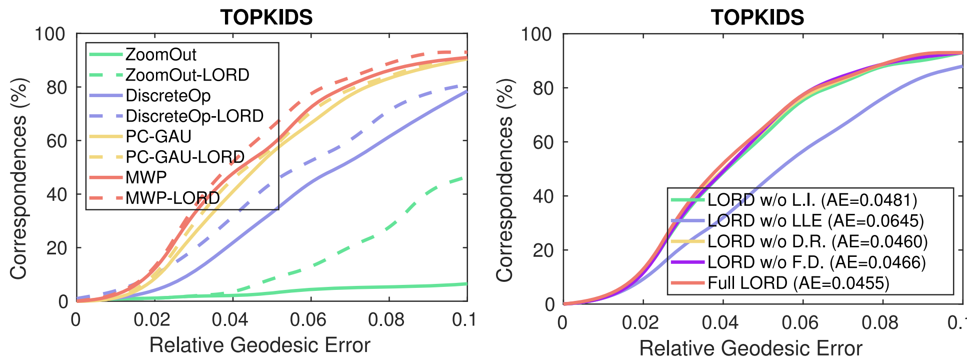

Fig. 12 Further analysis results. Left: generalizability verification using LORD on different function map methods, including ZoomOut, DiscreteOp, PC-GAU, and MWP. Right: an ablation study *w.r.t.* Locality preserving-based Initialization (L.I.), LLE-based refinement, Deformation-based Refinement (D.R.), and Functional Deformation (F.D.)

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Data Availability The data or code during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

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